DO TRONG KHANH

Software Engineer

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Ho Chi Minh City, Vietnam

Software engineer with 5+ years of experience building backend-heavy products: Kotlin and Spring Boot services, Go and Spark data pipelines, NestJS platforms and React frontends. I take features from spec to production (design, build, deploy and monitoring), with a focus on data integrity, resilient integrations and tests that cover failure modes, not just the happy path.

Experience

Spartan | Software Engineer — Mar 2025 – Present

Engineer embedded in US client product teams

LoanBud – small-business financing platform — Jan 2026 – Oct 2026

Kotlin, Micronaut, Java, Spring Boot, PostgreSQL, Redis, AWS SQS, React, TypeScript, TanStack, Terraform, Kubernetes

- Rebuilt the Hub-to-CRM data sync on a **transactional outbox**: Postgres triggers, `FOR UPDATE SKIP LOCKED` claims, exponential backoff computed in SQL and a horizontally scaled sync worker, replacing a polling job that silently dropped failed syncs. Added version-guarded upserts so out-of-order writes can't overwrite newer data.
- Built a **Meta Lead Ads ingestion pipeline** (signed webhook → SQS → Graph API) that is idempotent under duplicate delivery and recovers crashed consumers through leases; verified end to end on live traffic.
- Rebuilt the **CRM segment builder** across backend and frontend: application data exposed as typed, fail-closed JSONB segment fields, and a schema-driven filter UI validated by zod generated at runtime (two 7-PR stacks, ∼16k lines, dark-launched behind feature flags).
- Designed the **marketing attribution** model (write-once first-touch snapshots, insert-only touch history, legacy backfill) and the **Objective Context** API with a versioned snapshot contract for other services.
- Shipped **Twilio video meetings** with Google and Microsoft calendar sync and signed public join links, hardened with a Resilience4j circuit breaker, room quotas and Redisson-backed distributed rate limiting.
- Owned a rewrite of the legacy PHP core service in **Kotlin** (∼326 commits, +153k lines): behaviour-parity re-implementation, domain-manager split, a database-to-database migration, and the Terraform and secrets work.
- Also: per-field contact audit log, workflow auto-resume and safe editing of running workflows, credit-consent re-signing with PDF regeneration, partner CSV bulk import, and team tooling (Docker Compose, seed data, test split).

Dealops – CPQ software for complex pricing — Jul 2025 – Jan 2026

React, TypeScript, tRPC, Prisma, Tailwind CSS, Turborepo

- Built an in-app **AI chat assistant** from scratch (streaming responses, markdown rendering) and added segment filtering to the AI pricing assistant.
- Admin Dashboard: product-catalog management with tiered pricing, approval rules with an additional-logic builder, affinity-rule-aware product selection and segment filtering across the terms library.
- Quoting: the IC++ pricing model with 3-decimal payment-method fees, quote comparison, ramp revenue and an approval nudge flow (∼465 commits).

Deep Sky Climate – carbon removal project developer — Mar 2025 – Jul 2025

Go, Scala, Apache Spark, Apache Airflow, AWS Kinesis, EMR, DynamoDB, Kubernetes, Datadog

- **Founding engineer** on the IoT data platform: set the architecture, code conventions and CI/CD for three services.
- Go ingestion service: a custom **AWS Kinesis** consumer for Sparkplug B sensor telemetry with DynamoDB checkpointing and sharding, JWT/RBAC/OAuth auth, Datadog metrics, traces and profiling, and Kubernetes manifests.
- **Scala / Spark** batch jobs on EMR following the medallion architecture (ingestion, gold aggregations, dedup, customer egress), orchestrated by config-driven **Airflow** DAGs.

Also at Spartan: built and ran an internal Auth & Security lecture series (OIDC, service-to-service auth, XSS/CSRF with live attack sandboxes); buddy for new engineers and interns.

YouNet Media | Software Engineer — Mar 2022 – Mar 2025

Social listening and e-commerce analytics

- **EcomHeat** (cross-channel e-commerce analytics): APIs that export 1M+ rows to Google Sheets, cutting manual work by 90%; price-history monitoring with daily alerts and SKU management; Airflow and BigQuery jobs for brand metrics; a NestJS monorepo and Highcharts dashboards.
- **AppCore**: designed core NestJS packages (Database, Cache, Redis and Config modules) published to npm and used across 3+ projects; caching cut load times by ∼40%.
- **Lucene Search Service**: pre-filtered records with Apache Lucene before indexing into Solr, reducing query load.

QKIT Software | Backend Developer **Jan 2021 – Dec 2022**
Outsourcing software company

- **Military 7A Bidding**: designed the PostgreSQL schema and backend from scratch for tracking medical-supply bidding packages, with JWT authentication, role-based access and Swagger API docs.
- **QKIT Admin**: CMS for website content, blogs, clients and recruitment (NestJS, PostgreSQL, Prisma).

Skills

Languages: TypeScript, JavaScript, Kotlin, Java, Go, Scala, Python, SQL, C++
Backend: Node.js, NestJS, Micronaut, Spring Boot, tRPC, REST, GraphQL, RabbitMQ
Data: PostgreSQL, MySQL, MongoDB, Redis, DynamoDB, Apache Spark, Kafka, Airflow, Delta Lake
Frontend: React, TanStack Query and Form, zod, Tailwind CSS, Ant Design, Astro
Cloud & ops: AWS (SQS, Kinesis, EMR, S3, SageMaker), Docker, Kubernetes, Terraform, Datadog, GitHub Actions
Practices: transactional outbox, idempotent consumers, circuit breakers, rate limiting, feature-flagged releases, stacked PRs, AI-assisted development

Education

MSc in Information Systems | University of Information Technology, VNU-HCM **Dec 2024 – Present**
Thesis: a real-time mental-health state analysis system on Vietnamese social media (Kafka, Spark Streaming, Delta Lake, SageMaker); best model XLM-R at macro-F1 0.783.

BSc in Computer Science | University of Information Technology, VNU-HCM **Aug 2019 – Jun 2023**
GPA 8.5/10.

Projects & Certificates

karhdo.dev: personal blog and portfolio: Astro, Bun, Tailwind CSS, Drizzle on Neon Postgres, Vercel; this resume is built from LaTeX in the same repo.
Karhdo's Core: backend server packages for NestJS, published on npm.
TOEIC: 750/990.